

Hands-on introduction to (one) approach for evaluating informative hypotheses

Information criteria

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Example Palmer and Gough (P&G)

Palmer and Gough (2007) examined the difference in the attribution of importance of defective education as an explanation for criminal behavior

between three types of “offenders”:

 $(n_i = 20, 20, 31)$

ANOVA Example: Comparisons of 3 Means

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Palmer and Gough (2007) examined the difference in the attribution of importance of defective education as an explanation for criminal behavior between three types of “offenders”:
(1) non-offenders, (2) property offenders, and (3) person offenders.

ANOVA

$$y_j = \mu_1 d_{1j} + \mu_2 d_{2j} + \mu_3 d_{3j} + \epsilon_j,$$

where μ_i ($i = 1, 2, 3$) is the mean in group i ,
 d_{ij} is a dummy variable denoting group membership,
 ϵ_j is an error term and $\epsilon_j \sim N(0, \sigma^2)$.

$(n_i = 20, 20, 31)$

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$(n_i = 20, 20, 31)$

Hypotheses of interest

Example Palmer and Gough (P&G)

Palmer and Gough (2007) expect that

$$H_1: \mu_1 > \mu_2 > \mu_3.$$

Hypotheses of interest

Example Palmer and Gough (P&G)

Palmer and Gough (2007) expect that

- non-offenders (1) attribute more importance to defective education for explaining crime than the other two offenders (2 & 3):
i.e., $\mu_1 > \mu_2$ and $\mu_1 > \mu_3$,
- property offenders (2) attribute more importance to defective education than person offenders (3):
i.e., $\mu_2 > \mu_3$.

$$H_1: \mu_1 > \mu_2 > \mu_3.$$

Hypotheses of interest

Example Palmer and Gough (P&G)

Testing the null hypothesis

Example 3 means

Test H_0 with ANOVA F test:

$$H_0 : \quad \mu_1 = \mu_2 = \mu_3,$$

$$H_a : \quad \text{not } H_0.$$

Then, reject or not-reject ('accept') H_0 .

But, often not interested in H_0 !

Cannot say anything about $H_1 : \mu_1 > \mu_2 > \mu_3$.

NHST

Example Palmer and Gough

```
PandG_data <- read.table("Data_PalmerAndGough.txt",
                          header=TRUE)
PandG_data$group <- factor(PandG_data$group)
pairwise.t.test(PandG_data$Importance, PandG_data$group,
                p.adj = 'bonferroni')

##
## Pairwise comparisons using t tests with pooled SD
##
## data: PandG_data$Importance and PandG_data$group
##
##      1      2
## 2 0.191 -
## 3 0.011 1.000
##
## P value adjustment method: bonferroni
```


Confirmatory methods

Most researchers are able to specify “order-restricted” / “informative” / “theory-based” hypotheses, like $H_1 : \mu_1 > \mu_2 > \mu_3$.
Use prior knowledge and/or expertise in hypothesis.

Note: 'model' refers to hypothesis.

GORIC

Example Palmer and Gough, that is, 3 means

$$H_0 : \mu_1 = \mu_2 = \mu_3,$$

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

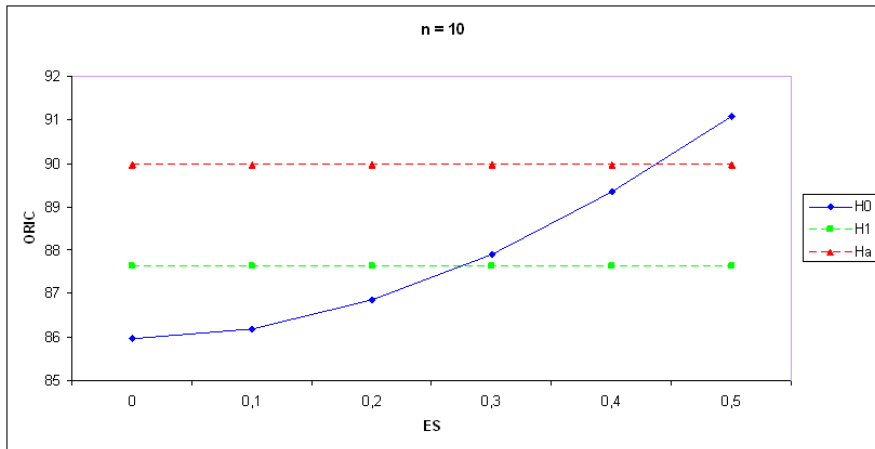
$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC – will be explained later

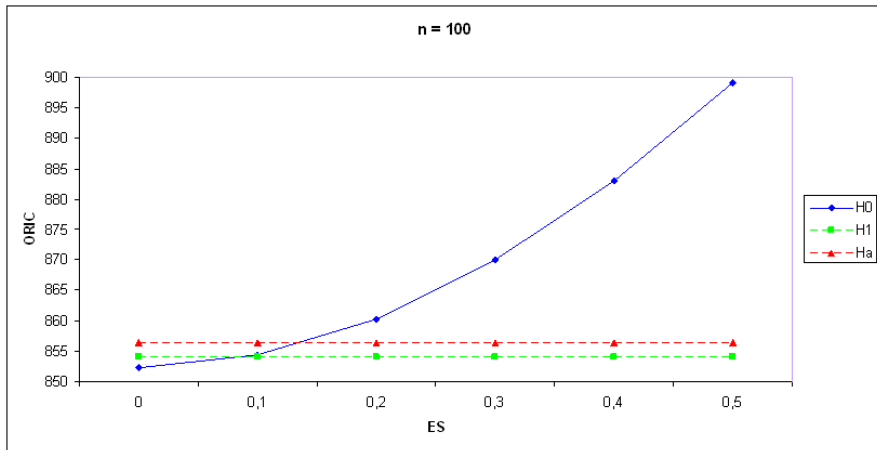
Model	Fit	Complexity	GORIC	GORIC weights
H_0	-196.36	2.00	396.71	0.02
H_1	-191.89	2.81	389.41	0.75
H_u	-191.89	4.00	391.79	0.23

Confirmatory methods (e.g., GORIC) have more “power” than their exploratory counterparts (e.g., AIC; cf. one- vs two-sided testing).

Confirmation more power: 1 data set. GORIC values for 3 groups, effect size ES , and $n = 10$ observations per group



Confirmation more power: 1 data set. GORIC values for 3 groups, effect size ES , and $n = 100$ observations per group



ANOVA Example: Comparisons of 3 Means

Simple, hypothetical example

Examine the difference in happiness

between three types of “treatments”:

(A) treatment A, (B) treatment B, and (C) no treatment / control.

Possible (sets of) theory-based hypothesis/-es:

$$H_1 : \quad \mu_A > \mu_B > \mu_C.$$

$$H_1 : \quad \mu_A > \mu_B > \mu_C,$$

$$H_2 : \quad \mu_B > \mu_A > \mu_C.$$

$$H_1 : \quad \{\mu_A, \mu_B\} > \mu_C.$$

$$H_1 : \quad \mu_A - \mu_B > 0.1 \quad \& \quad \mu_B - \mu_C > 0.2.$$

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GORIC weights

Examples with R code

GORICA

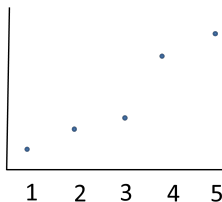
GORIC(A) in JASP

End & Extra

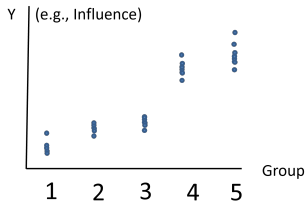
Intermezzo: Balance Fit and Complexity (1/6)

Example 5 means

Data for 5 groups – Oversimplified representation

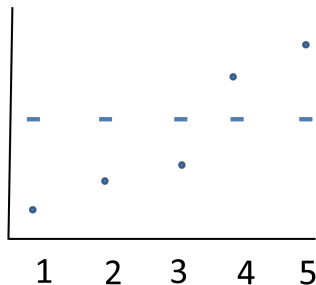


Data for 5 groups – Better representation (scatter of data points)



Intermezzo: Balance Fit and Complexity (2/6)

Example 5 means

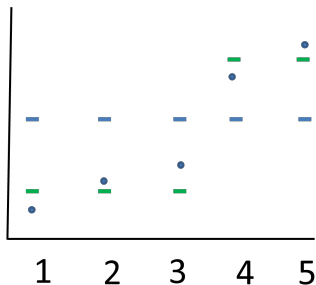


$$\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu$$

parameters: 1 mean parameter (i.e., μ)
Fit: Bad fit

Intermezzo: Balance Fit and Complexity (3/6)

Example 5 means



$$\mu_1 = \mu_2 = \mu_3 = \mu_{\text{low}}$$

$$\mu_4 = \mu_5 = \mu_{\text{high}}$$

parameters:

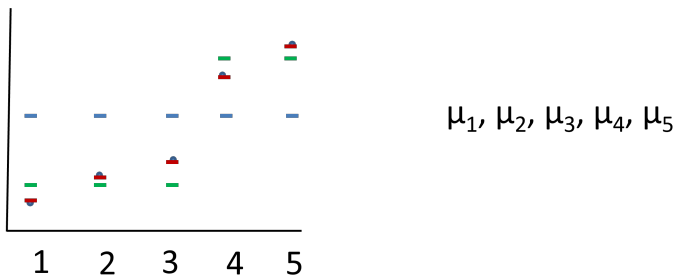
2 mean parameters (i.e., μ_{low} and μ_{high})

Fit:

Better than with one parameter, even good fit.

Intermezzo: Balance Fit and Complexity (4/6)

Example 5 means



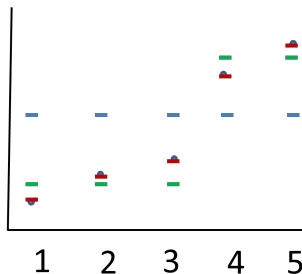
parameters: 5 mean parameter (complex)

Fit: Better than with two parameters, even best possible fit.

Thus: Best fit (= highest likelihood), but also most complex (= highest penalty).

Intermezzo: Balance Fit and Complexity (5/6)

Example 5 means



$$\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu$$

$$\mu_1 = \mu_2 = \mu_3 = \mu_{\text{low}} \text{ \& }$$

$$\mu_4 = \mu_5 = \mu_{\text{high}}$$

$$\mu_1, \mu_2, \mu_3, \mu_4, \mu_5$$

In this example: **Two means** may be best trade-off between fit & complexity.

Intermezzo: Balance Fit and Complexity (6a/6)

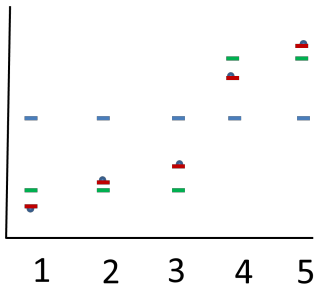
Order-restrictions

Previous slides demonstrated balancing fit and complexity for equality ('=') restrictions.

For inequality/order ('<', '>', and possible some '=') restrictions, the calculation of the fit and complexity is more complex :-), but the idea of balancing is the same.

Intermezzo: Balance Fit and Complexity (6b/6)

Order-restrictions: From “Example 5 means” to ‘Example 2 means’



In theory-based hypotheses, we also incorporate **order-restrictions** (e.g., $\mu_1 < \mu_2$).
Then, helpful to look at likelihood using **contour plots** (not scatter plot, as done here).

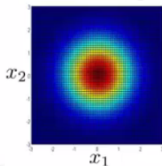
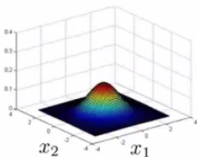
For ease, I will next use **2 means** (instead of 5).

Intermezzo: Contour plot (2/2)

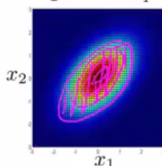
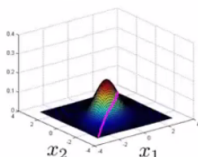
Example 2 means

Multivariate Gaussian (Normal) examples

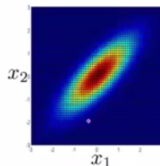
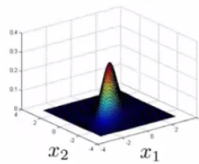
$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$



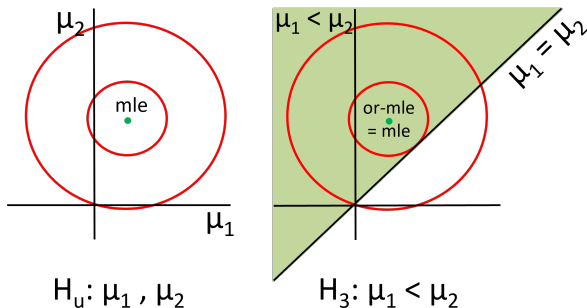
$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$$



Andreas

Note: if in agreement, or-mle = mle

Example 2 means with or-mle = mle



mle in H_3 : highest likelihood is in allowable (= green) space.
Hence, or-mle = mle.

Idea fit

Example Palmer and Gough, that is, 3 means

$$H_0 : \mu_1 = \mu_2 = \mu_3,$$

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_1 > \mu_2 < \mu_3,$$

$$H_3 : \mu_1 < \mu_2 < \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

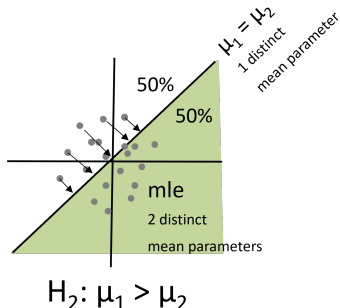
```
## group1 group2 group3
## 11.95 9.75 8.77
```

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_0	-196.36	2.00	396.71	0.01
H_1	-191.89	2.81	389.41	0.56
H_2	-192.34	3.19	391.05	0.25
H_3	-196.36	2.81	398.34	0.01
H_u	-191.89	4.00	391.79	0.17

Idea complexity: penalty (PT)

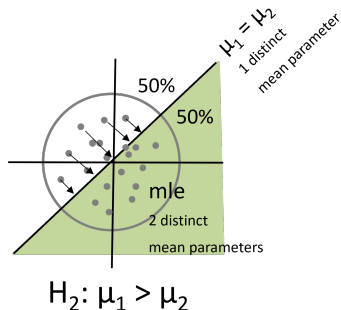
Example 2 means



Note: sampling from distribution with $\mu_1 = \mu_2 = 0$.
 Complexity H_2 is $PT_2 = 1 + 0.5 \times 1 + 0.5 \times 2 = 2.5$.
 PT = number of expected distinct/free parameters.

Idea complexity: penalty (PT)

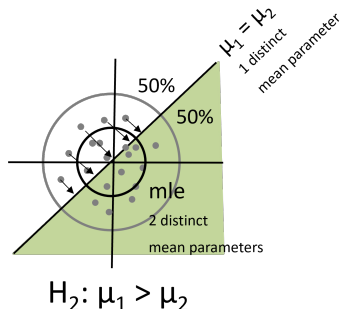
Example 2 means: Contour distribution



Sampling from distribution with $\mu_1 = \mu_2 = 0$. (Note: not likelihood!)
 Contour denotes distribution for a variance value.
 Choice variance does not matter, see next slide.

Idea complexity: penalty (PT)

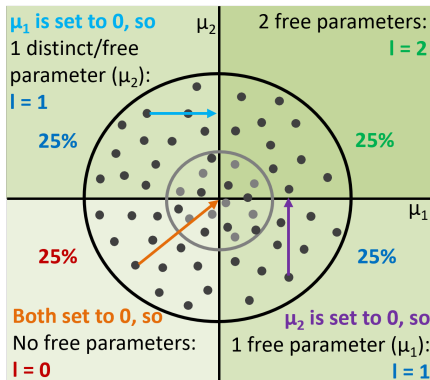
Example 2 means: PT value independent from choice variance



For both variance values, that is, for both contours:
Complexity H_2 is $PT_2 = 1 + 0.5 \times 1 + 0.5 \times 2 = 2.5$.
PT = number of expected distinct/free parameters.

Extra: Levels and level probabilities

Example 2 means



$H_4: \mu_1 > 0, \mu_2 > 0$

Level (l): number of distinct/free parameters.

Level probability (LP_l): probability that there are l levels.

$$PT_4 = 1 + 0.25 \times 0 + (0.25 + 0.25) \times 1 + 0.25 \times 2 = 2.$$

Idea complexity

Example Palmer and Gough, that is, 3 means

$$H_0 : \mu_1 = \mu_2 = \mu_3,$$

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_1 > \mu_2 < \mu_3,$$

$$H_3 : \mu_1 < \mu_2 < \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

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GORIC: Lowest value is best

GORIC is like AIC expected distance from the truth (KL-distance).
One wants smallest distance.

Or: One wants the smallest 'misfit + complexity'.

Note: $\text{GORIC} = -2 \text{ fit} + 2 \text{ complexity} = 2 \text{ misfit} + 2 \text{ complexity}$

Hence, smallest value is best.

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Hence, smallest value is best.

GORIC

Example Palmer and Gough

- H_0 : $\mu_1 = \mu_2 = \mu_3$ (include only when of interest)
- H_1 : $\mu_1 > \mu_2 > \mu_3$
- H_2 : $\mu_2 > \mu_1 > \mu_3$ (if competing theory/hypothesis)
- H_u : μ_1, μ_2, μ_3 .

GORIC

Model	Fit	Complexity	GORIC
H_0	-196.36	2.00	396.71
H_1	-191.89	2.81	389.41
H_2	-193.70	2.81	393.03
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Hands-on/Demo (1a): GORIC

- Go to <https://github.com/rebeccakuiper/Tutorials>:
 - Click on green button called Code.
 - Download zip (last option in list).
 - Unzip it on your machine (that folder is now your working dir.).
- Start Rstudio. Optional: make project.
- Open 'Tutorial_GORIC_restriktor_ANOVA.html', 'PalmerAndGough_and_Lucas.R', and/or '**Hands-on_1_GORIC_Unc_ANOVA_restriktor.R**' ('Hands-on files').
- Install packages and load them.
- Read and inspect data.
Use **Data_PalmerAndGough.txt** and/or Data_Lucas.txt.
- Run model (lm()).
- Specify hypotheses (make up your own).
Note: Use names used in the model.
- Run goric().
- Inspect and interpret output.
Note: 'GORIC weights' will be explained next.



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End & Extra

Illustration of the GORIC weights (w_m)

Example Palmer and Gough

$H_0 : \mu_1 = \mu_2 = \mu_3$ (include only when of interest)

$$H_1: \mu_1 > \mu_2 > \mu_3$$
$$H_2: \mu_2 > \mu_1 > \mu_3 \text{ (if competing theory/hypothesis)}$$
$$H_\mu : \quad \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.68
H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

H_1 is $0.68 / 0.11 \approx 6.11$ times more supported than H_2 .

- If needed: Start Rstudio again (optional: make project) and then also load packages again.
- Read and inspect data.
- Run model (`lm()`).
- Specify hypotheses (make up your own).
Note: Use names used in the model.
- Run `goric()`.
- Inspect and interpret output: Focus on GORIC weights.

GORIC output and interpretation

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.68
H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

H_1 and H_2 are not weak (nl, better than H_u - will be explained in a bit).

H_1 is $0.68 / 0.11 \approx 6.11$ times more supported than H_2 .

Another illustration: one-way ANOVA

Lucas: 5 groups

Lucas (2003) investigated difference between female and male leadership w.r.t. influence of the leader.

Five experimental groups:

1. a randomly selected male leader
2. a randomly selected female leader
3. male leader selected via task
4. female leader selected via task
5. female leader selected via task + institutionalized female leadership via movie

(Two informative) hypotheses of interest

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 \text{ (include only when of interest)}$$
$$H_1 : \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$$
$$H_2 : \mu_3 > \mu_1 > \mu_4 = \mu_5 > \mu_2$$
 $H_u : \mu_1, \mu_2, \mu_3, \mu_4, \mu_5$ (included as failsafe)

(Two informative) hypotheses of interest

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 \text{ (include only when of interest)}$$
$$H_1 : \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$$
$$H_2 : \mu_3 > \mu_1 > \mu_4 = \mu_5 > \mu_2$$
 $H_u : \mu_1, \mu_2, \mu_3, \mu_4, \mu_5$ (included as failsafe)

Descriptive statistics of Lucas' Data

Group	Mean Influence	SD	n
1 (male, random)	2.33	1.86	30
2 (female, random)	1.33	1.15	30
3 (male, selected)	3.20	1.79	30
4 (female, selected)	2.23	1.45	30
5 (female, selected+)	3.23	1.50	30

GORIC weights (w_m)

Lucas

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-278.05	3.46	563.02	0.90
H_2	-281.76	3.14	569.79	0.03
H_u	-278.05	6.00	568.10	0.07

H_1 is $0.9 / 0.03 \approx 29.59$ times more supported than H_2 .
Thus, there is quite some evidence that H_1 is the best of this set.

Note:

H_1 is $0.9 / 0.07 > 1$ times more supported than H_u .
Hence, H_1 not weak and thus comparable to the other
(weak and non-weak) informative hypotheses (as explained in a bit).

The failsafe hypothesis will be preferred over the informative hypotheses in the set, if the sample size is large enough.

E.g.: $w_1 = .08$, $w_2 = .02$, and $w_u = .90$.

Use of H_u

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.68
H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

If at least one informative hypothesis not weak (i.e., $w_m > w_u$ or $w_m/w_u > 1$), then compare informative hypotheses.

Hence: H_u is only a failsafe, not another hypothesis of interest.

Use of H_u

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

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Hence: H_u is only a failsafe, not another hypothesis of interest.

Use of H_u

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights	without Hu
H_1	-191.89	2.81	389.41	0.68	0.86
H_2	-193.70	2.81	393.03	0.11	0.14
H_u	-191.89	4.00	391.79	0.21	NA

H_1 is not weak and is $0.68 / 0.11 = 0.86 / 0.14 \approx 6.11$ times more supported than competing hypothesis H_2 .

H_1 vs H_u

Palmer & Gough

What if only one informative hypothesis:

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.77
H_u	-191.89	4.00	391.79	0.23

H_1 is $0.77 / 0.23 \approx 3.27$ times more supported than H_u .

BUT: H_u includes H_1 .

So, support for H_u contains support for H_1 .

H_1 vs H_u

Palmer & Gough

What if only one informative hypothesis:

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.77
H_u	-191.89	4.00	391.79	0.23

H_1 is $0.77 / 0.23 \approx 3.27$ times more supported than H_u .

BUT: H_u includes H_1 .

So, support for H_u contains support for H_1 .

H_1 vs H_u

Palmer & Gough

What if only one informative hypothesis:

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.77
H_u	-191.89	4.00	391.79	0.23

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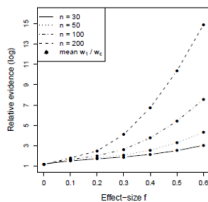
BUT: H_u includes H_1 .

So, support for H_u contains support for H_1 .

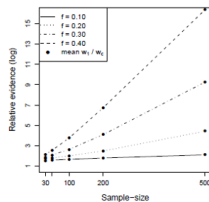
Alternative failsafe: Complement of H_m

Based on simulation

vs complement

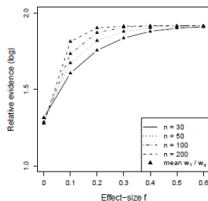


(a)

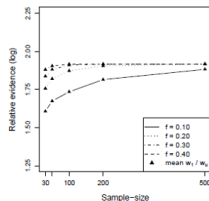


(b)

vs unconstrained



(c)



(d)

R code: complement

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_c : \text{not } H_1.$$

GORIC

```

H1 <- 'group1 > group2 > group3'
# vs its complement (default in case of one hypothesis)
#
# GORIC (using goric function in restriktor package)
library(restriktor)
set.seed(123) # Set seed value
goric.PandG_1c <- goric(fit.PandG,
                        hypotheses = list(H1))
    
```


- If multiple informative, theory-based hypotheses:
Goal: Comparison of informative, theory-based hypotheses.
Safeguard: Unconstrained.
- If one informative, theory-based hypothesis (H_m):
Goal: Compare H_m with all other possibilities / theories.
Safeguard: Its complement.

Note on use of safeguard hypotheses (2/2)

My advise and default in goric function:

- If multiple informative, theory-based hypotheses:
Goal: Comparison of informative, theory-based hypotheses.
Safeguard: Unconstrained.
- If one informative, theory-based hypothesis (H_m):
Goal: Compare H_m with all other possibilities / theories.
Safeguard: Its complement.

Can overrule by using argument 'comparison'

(comparison = "unconstrained", "complement", or "none").

Only use 'comparison = "none"' when your set contains all possibilities (e.g., $\beta > 0$ & $\beta < 0$).

Extra: complement & 'Heq' (4/4)

Note

H_{eq} is only one boundary hypothesis, and the most strict one.

I would only include H_{eq} when you a priori want to evaluate this (possibly to rule it out).

When the truth lies on the border of H_1 and H_c , then the loglik.weights are approximate equal.

If you find this, then I would go into an exploration phase (and evaluate all (or some) boundary hypotheses).

Hands-on/Demo (2): GORIC - H_m vs complement

Let's practice.

- If needed: Start Rstudio again (optional: make project) and then also load packages again.
- Optional: Open 'Tutorial_GORIC_restriktor_ANOVA.html', 'PalmerAndGough_and_Lucas.R', and/or '**Hands-on_2_GORIC_Compl_ANOVA_restriktor.R**'.
- Read and inspect data.
Use Data_PalmerAndGough.txt and/or **Data_Lucas.txt**.
- Run model (lm()).
- Specify hypotheses (make up your own). E.g., for Lucas:

$$H_1 : \quad \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$$

Note: Use names used in the model.

- Run goric()
- Inspect and interpret output.

GORICA

GORIC: Normal linear models

GORIC can easily be applied to normal linear models (e.g., ANOVA models or regression models).

GORIC: Other statistical models

In case of other statistical models (e.g., a SEM model), more cumbersome to calculate maximized order-restricted log likelihood and thus GORIC.

GORICA: All statistical models

Therefore, GORICA: asymptotic expression for GORIC.
Can be used for all types of statistical models.

Reference:

Altınışık, Y., Van Lissa, C. J., Hoijtink, H., Oldehinkel, A. J., and Kuiper, R. M. (2021). Evaluation of inequality constrained hypotheses using a generalization of the AIC. *Psychological Methods*, 26(5), 599–621.

<https://doi.org/10.1037/met0000406>

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<https://doi.org/10.1037/met0000406>

GORICA

Similarities with GORIC

- Form: $GORICA_m = -2 \text{ fit} + 2 \text{ complexity}$.
- Broad type of restrictions.

Differences compared to GORIC

- Uses asymptotic expression of the likelihood (is a normal):
can therefore be easily applied to all types of statistical models.
Disadvantage: might work less well in case of small samples.
- Does not need data set; mle's and their covariance matrix suffice.
- Can leave out nuisance parameters (i.e., not part of hypotheses).

Note

In case of normal linear models or, otherwise, for not too small samples:
GORICA weights = GORIC weights.

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In case of normal linear models or, otherwise, for not too small samples:
GORICA weights = GORIC weights.

R code: GORICA

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_c : \text{not } H_1.$$

GORICA: type = "gorica" (not default for lm objects)

```

H1 <- 'group1 > group2 > group3'
# vs its complement (default in case of one hypothesis)
#
# GORICA (using goric function in restriktor package)
library(restriktor)
set.seed(123) # Set seed value
gorica.PandG_C <- goric(fit.PandG,
                        hypotheses = list(H1),
                        type = "gorica") # needed if lm object
    
```

GORICA

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_c : \text{not } H_1.$$

GORIC

Model	Fit	Complexity	GORICA	GORICA weights
H1	-1.90	1.81	7.42	0.79
complement	-2.34	2.69	10.06	0.21

H_1 is $0.79 / 0.21 \approx 3.74$ times more supported than its complement, that is, any other hypothesis.

Note: GORIC(A) weights are the same (ratio may differ a bit).

GORICA

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

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GORIC

Model	Fit	Complexity	GORICA	GORICA weights
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H_1 is $0.79 / 0.21 \approx 3.74$ times more supported than its complement, that is, any other hypothesis.

Note: GORIC(A) weights are the same (ratio may differ a bit).

Hands-on/Demo (3): GORICA

Let's practice.

- If needed: Start Rstudio again (optional: make project) and then also load packages again.
- Optional: Open 'Tutorial_GORIC_restriktor_ANOVA.html', 'PalmerAndGough_and_Lucas.R', and/or '**Hands-on_3_GORICA_UncAndCompl_ANOVA_restriktor.R**'.
- Read and inspect data.
Use Data_PalmerAndGough.txt and/or **Data_Lucas.txt**.
- Run model (lm()).
- Specify hypotheses (make up your own). E.g., for Lucas:

$$H_1 : \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$$

Note: Use names used in the model (or overwrite those).

- Run goric(); now, add (since lm object):
`type = "gorica"`
- Inspect and interpret output.

Power improvement

Power improvement via

- Using order restrictions
(cf. one-sided vs two-sided testing).
- Having limited set of hypotheses.
Note: Null hypothesis can be one of them, but only if of interest!
Try to think outside of the null-hypothesis box :-).
- (if:) Specifying restrictions for multiple outcomes within each informative hypothesis.

If you do not have a priori expectations and sample size is large enough:

- 1) You could use part of your data to generate one or more hypotheses.
- 2) This/These can then be evaluated in the other part of the data.

GORIC in JASP: screenshot

Palmer & Gough

Data_PalmerAndGough_JASP
(C:\WER\Dev - 503 - winter - jasp en R Lab\JASP labmeeting)

Descriptives

T-Tests

ANOVA

Mixed Models

Regression

Frequencies

Factor

Bain

SEM

R (Beta)

▼ Order Restricted Hypotheses

Enter each restriction of one hypothesis on a new line, e.g.,
factorLow == factorMid
factorMid < factorHigh
where 'factor' is the factor (or covariate) name and 'Low'/'Mid'/'High' are the factor level names.
Click on the 'plus' icon to add more hypotheses.
Click the information icon for more examples.

Syntax settings

☐ Include intercept
☐ Show available coefficients

Set for all models

☐ Model summary
☐ Marginal means
☐ Informed hypothesis tests

Model 1

+

group1 > group2 > group3

Ctrl + Enter to apply. Click on the blue button above for help on the restriction syntax

☐ Summary for Model 1
☐ Marginal means for Model 1
☐ Informed hypothesis tests for Model 1

Results

ANOVA

ANOVA - Importance

Cases	Sum of Squares	df	Mean Square	F	p
group	123.960	2	61.980	4.554	0.014
Residuals	925.418	68	13.609		

Note. Type III Sum of Squares

Order Restricted Hypotheses

Model Comparison

Model Comparison Table

Model	Log-likelihood	Penalty	GORIC	Weight	Weights ratio
Model 1	-191.893	2.814	389.415	0.789	3.731
Complement	-192.338	3.686	392.048	0.211	1.000

Note. Weights ratios indicate the relative weight for each model against the "complement" model. GORIC = Generalized Order-Restricted Information Criterion (Kuiper, Hoijtink, & Silvapulle, 2011).

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GORIC and GORICA in JASP

	GORIC R	GORIC JASP	GORICA R	GORICA JASP
ANOVA	✓	✓	✓	
RM-ANOVA			✓	✓
Regression	✓	Not yet	✓	
SEM			✓	Future?
Other model			✓	Future?

GORIC and GORICA in JASP

GORIC or GORICA in the following statistical models (as part of that model)

The image displays three JASP statistical model configuration windows, each with a list of options on the left and a summary table on the right.

- ANOVA Window:** The left sidebar has "Order Restricted Hypotheses" circled in blue. The right panel shows a table with columns: Cases, Sum of Squares, df, Mean Square, F, and p. The title "ANOVA" is circled in red.
- ANCOVA Window:** The left sidebar has "Order Restricted Hypotheses" circled in blue. The right panel shows a table with columns: Cases, Sum of Squares, df, Mean Square, F, and p. The title "ANCOVA" is circled in red. A separate panel on the right shows a list of variables with "id" selected.
- Repeated Measures ANOVA Window:** The left sidebar has "Order Restricted Hypotheses" circled in blue. The right panel shows a table with columns: Cases, Sum of Squares, df, Mean Square, F, and p. The title "Repeated Measures ANOVA" is circled in red. A separate panel on the right shows a list of variables with "id" selected.

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Example Null hypothesis (H_0) vs Informative hypothesis (H_m)

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Failsafe/Safeguard hypotheses: Unconstrained & Complement

GORICA

GORIC(A) in JASP

End & Extra

The End

GORIC(A)

Thanks for listening!

Are there any questions?

Websites

<https://github.com/rebeccakuiper/Tutorials>

www.uu.nl/staff/RMKuiper/Extra1

www.uu.nl/staff/RMKuiper/Extra2

informative-hypotheses.sites.uu.nl/software/goric/

E-mail

r.m.kuiper@uu.nl

Notes

Note on type of hypotheses: Magnitude-hypotheses

Hypotheses can regard height of effect size.

In the case of Cohen's d , one could evaluate:

$$H_1 : d < 0,$$

$$H_2 : d > 0,$$

$$H_3 : d > 0.2,$$

$$H_4 : d > 0.5,$$

$$H_5 : d > 0.8.$$

$$H_1 : d < 0,$$

$$H_2 : 0 < d < 0.2,$$

$$H_3 : 0.2 < d < 0.5,$$

$$H_4 : 0.5 < d < 0.8,$$

$$H_5 : d > 0.8.$$

Now, overlapping hypotheses.

Now, range restrictions
(complexity as if equalities).

Or better: One of these versus its complement.

Or: incorporate such cut-off values when inspecting ordering of parameters (say, means; e.g., $\mu_1 - \mu_2 > 0.2 \times s.d.$).

Note on possibilities multiple studies

- Update GORIC(A) values & weights.
More data collected: (re-)calculate.
- Update hypotheses.
If no expectations (or no support), use exploration and confirmation:
First data set (or a part of it) generates one or more hypotheses.
Other data set (or part) used to determine evidence / support.
Download 'Tutorial_GORIC_restriktor_UpdateHypo.html' and/or
'Hands-on_4_GORIC_UpdateHypo_restriktor.R' from
<https://github.com/rebeccakuiper/Tutorials>.
- Aggregate evidence for hypotheses.
Aggregate the support for theories (diverse designs allowed).
Bear in mind: Meta-analysis aggregates parameter estimates or
effect sizes which need to be comparable (often same designs
required).
Download 'Tutorial_GORIC_restriktor_evSyn.html' and/or
'Hands-on_4_GORIC_evSyn_restriktor.R' from
<https://github.com/rebeccakuiper/Tutorials>.

Note: On github site, go to Code (green button) and download zip.

Extra material (1/2)

- GORICA on logistic Regression Model
 - Article: doi.org/10.1037/met0000406
- GORICA on SEM
 - Article: www.tandfonline.com/doi/full/10.1080/10705511.2020.1836967.
 - R scripts: github.com/rebeccakuiper/GORICA_in_SEM.
- GORICA on cross-lagged panel model (CLPM)
 - Article: doi.org/10.1111/bjep.12455.
 - R scripts: github.com/rebeccakuiper/GORICA_in_CLPM.
- GORICA on Random-Intercept CLPM (RI-CLPM)
 - Article: Sukpan, C. and Kuiper, R.M. (2023). How to evaluate causal dominance hypotheses in lagged effects models.
 - R scripts: github.com/rebeccakuiper/GORICA%20in%20RI-CLPM.
- GORICA for EffectLiteR
 - R scripts: <https://github.com/rebeccakuiper/Tutorials/tree/main/GORICA%20for%20EffectLiteR>.

Note: On github site, go to Code (green button) and download zip.

Note on conditional error probability (1/3)

using PMPs and GORIC(A) weights

H_m	weights
H_1 : Sex Match	.04
H_2 : Gender Role Match	.81
H_3 : Sex Mismatch	.01
H_4 : Gender Role Mismatch	.00
H_u :	.14

The conditional error probability for preferred hypotheses H_2 is $1 - .81 = .19$.

Note on conditional error probability (3/3)

using PMPS and GORIC(A) weights

What does conditional error probability reflect when hypotheses overlap? Notably, when hypotheses overlap they can also share/divide support (like H_u does with any hypothesis).

My advise:

Only use error probabilities if one hypothesis versus its complement.
(Or when you are sure that there is no overlap in hypotheses;
and they preferably do cover all possibilities)

H_m	weights
H_2 : Gender Role Match	.82
H_c : complement H_2	.18

Simulation study

Results Confirmation ($k = 3$ and $n = 50$)

$$H_{30C} : \mu_1 = \mu_2 = \mu_3$$

$$H_{31C} : \mu_1 < \mu_2 < \mu_3$$

$$H_{32C} : \mu_1 = \mu_2 < \mu_3$$

$$H_{33C} : \mu_1 < \mu_2 > \mu_3$$

$$H_{34C} : \mu_1, \mu_2, \mu_3$$

1 – Type I error (here):

preferring the true null hypothesis,
given the set of hypotheses (!).

'ES'	Method	'Prior'	H_{30C}	H_{31C}	H_{32C}	H_{33C}	H_{34C}
0	ORIC		0.662	0.065	0.119	0.120	0.034
0	BMS	3	0.865	0.012	0.075	0.036	0.012
0	BMS	2	0.774	0.022	0.124	0.058	0.022
0	BMS	1	0.656	0.047	0.147	0.120	0.033
2	ORIC		0.148	0.506	0.311	0.029	0.006
2	BMS	3	0.299	0.299	0.371	0.031	0.001
2	BMS	2	0.252	0.328	0.381	0.040	0.000
2	BMS	1	0.125	0.494	0.321	0.063	0.000
5	ORIC		0.000	0.955	0.044	0.001	0.000
5	BMS	3	0.000	0.887	0.112	0.001	0.000
5	BMS	2	0.001	0.890	0.107	0.002	0.000
5	BMS	1	0.000	0.928	0.070	0.002	0.000

Results Confirmation ($k = 3$ and $n = 50$)

$$H_{30C} : \mu_1 = \mu_2 = \mu_3$$

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$$H_{34C} : \mu_1, \mu_2, \mu_3$$

1 – Type II error (here):
 preferring the true hypothesis,
 given the set of hypotheses (!).

'ES'	Method	'Prior'	H_{30C}	H_{31C}	H_{32C}	H_{33C}	H_{34C}
0	ORIC		0.662	0.065	0.119	0.120	0.034
0	BMS	3	0.865	0.012	0.075	0.036	0.012
0	BMS	2	0.774	0.022	0.124	0.058	0.022
0	BMS	1	0.656	0.047	0.147	0.120	0.033
2	ORIC		0.148	0.506	0.311	0.029	0.006
2	BMS	3	0.299	0.299	0.371	0.031	0.001
2	BMS	2	0.252	0.328	0.381	0.040	0.000
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5	BMS	2	0.001	0.890	0.107	0.002	0.000
5	BMS	1	0.000	0.928	0.070	0.002	0.000

Extra: Prior choice for BMS

Prior

$$P(\mu, \sigma^2) = P(\mu_1) \times \dots \times P(\mu_k) \times P(\sigma^2),$$

where $P(\mu_i) = P(\mu) \forall i = 1, \dots, k$.

$P(\mu)$ is a data-based normal distribution, with parameters β_0 and τ_0^2 , and $P(\sigma^2)$ an inverse chi-squared distribution (see Klugkist et al., 2005)

β_0 and τ_0^2 data-based hyperparameters

For each μ_i ($i = 1, \dots, k$) a credibility interval is computed:

estimated group mean $\pm$ $PV \times$ *the group standard error*.

The credibility interval for μ_i is represented by " $[LB_i, UB_i]$ ".

Lowest lower bound: LB_{min} .

Highest upper bound: UB_{max} .

Then, $\beta_0 = \frac{LB_{min} + UB_{max}}{2}$ and $\tau_0^2 = \left[\frac{UB_{max} - LB_{min}}{2} \right]^2$.

Three types of prior vagueness: BMS PV 1, BMS PV 2 and BMS PV 3